LECTURE 4

ALGEBRIC & PROBABILISTIC VIEW OF DATA

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Basic symbols and terminology

Vectors and matrices

$$\vec{x} \quad or \quad x$$

$$\text{If} \quad \vec{x} = \begin{pmatrix} 3 \\ 6 \\ 8 \end{pmatrix} \quad \text{then} \quad x_1 = 3$$

```
x = [3, 6, 8]
import numpy as np
x = np.array([3, 6, 8])
```

Vectors and Matrices

	Office 1	Office 2	Office 3
HR	57	67	65
Engineering	89	87	98
Management	94	84	60

$$X = \begin{pmatrix} 57 & 67 & 65 \\ 89 & 87 & 98 \\ 94 & 84 & 60 \end{pmatrix}$$

Any other examples where matrices are used in ML or DL?

Summation

X = [1, 2, 3, 4, 5]

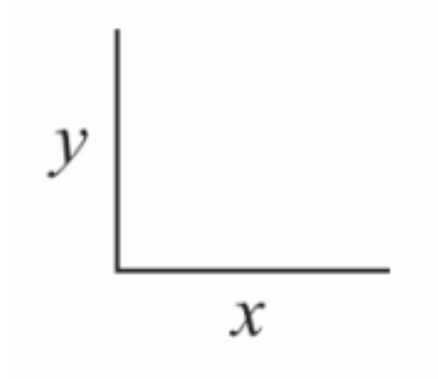
$$\sum x_i =$$

$$\sum x_i = 15$$

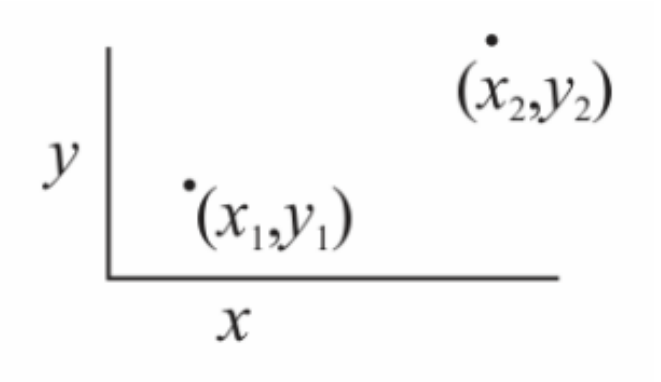
Dot Product

$$\begin{pmatrix} 3 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 9 \\ 5 \end{pmatrix} = 3 * 9 + 7 * 5 = 62$$

Graphs



Graphs



The **slope** between these two points is defined as follows:

$$slope = m = \frac{y_2 - y_1}{x_2 - x_1}$$

Logarithms/exponents

exponent

$$2^{4} = 2 \cdot 2 \cdot 2 \cdot 2 = 16$$

base

Logarithms/exponents

- $\log_3 81 = 4$ because $3^4 = 81$
- $\log_5 125 = 3$ because $5^3 = 125$

Set Theory

Homework

Linear Algebra Revision

Matrix: Rectangular array of numbers:

$$\begin{bmatrix} 1402 & 191 \\ 1371 & 821 \\ 949 & 1437 \\ 147 & 1448 \end{bmatrix}$$

4 x 2 matrix

$$\mathbb{R}^{4\times 2}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

2

3

2 x 3 matrix

Dimension of matrix: number of rows x number of columns

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Linear Algebra Revision

Matrix Elements (entries of a matrix)

An n x 1 matrix is called a vector.

Matrix Elements (entries of matrix)

$$A = \begin{bmatrix} 1402 & 191 \\ 1371 & 821 \\ 949 & 1437 \\ 147 & 1448 \end{bmatrix}$$

$A_{ij} =$ "i, j entry" in the ith row, j^{th} column.

$$y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

Linear Algebra Revision

$$\begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 0.5 \\ 2 & 5 \\ 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 0.5 \\ 2 & 5 \end{bmatrix} =$$

Linear Algebra Revision

$$\begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 0.5 \\ 2 & 5 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0.5 \\ 4 & 10 \\ 3 & 2 \end{bmatrix}$$

$$\underbrace{\cancel{\begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix}}}_{3\times 2} + \underbrace{\cancel{\begin{bmatrix} 4 & 0.5 \\ 2 & 5 \end{bmatrix}}}_{2\times 2} = \text{error}$$

Linear Algebra Revision

$$3 \times \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} /4 =$$

Linear Algebra Revision

$$3 \times \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 6 & 15 \\ 9 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} /4 = \begin{bmatrix} 1 & 0 \\ \frac{3}{2} & \frac{3}{4} \end{bmatrix}$$

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Linear Algebra Revision

$$\begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 3 & 0 & 4 \\ -1 & -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 1 \end{bmatrix} =$$

Linear Algebra Revision

$$\begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 3 & 0 & 4 \\ -1 & -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \\ -7 \end{bmatrix}$$

Linear Algebra Revision

$$\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 5 & 2 \end{bmatrix} =$$

Linear Algebra Revision

Matrix operations are NOT commutative in nature.

- A x B != B x A

Matrix operations are associative in nature.

- A x (B x C) == (A x B) xC

Linear Algebra Revision (Identity Matrix)

For any matrix A,

$A.I = I.A = A$

Denoted I (or $I_{n \times n}$).

Examples of identity matrices:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 x 2 3 x 3 4 x 4

Linear Algebra Revision

Inverse Matrix

- If A is an m*m matrix its inverse will be:
- $AA^{-1} = A^{-1}A = I$

$$\begin{bmatrix} 3 & 4 \\ 2 & 16 \end{bmatrix} \begin{bmatrix} 0.4 & -0.1 \\ -0.05 & 0.075 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2\times 2}$$

Linear Algebra Revision

Matrix Transpose:

- Obtained by flipping the matrix around the axis to

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 5 & 9 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 0 & 9 \end{bmatrix}$$

Let A be an m x n matrix, and let $B = A^T$.
Then B is an n x m matrix, and

$$B_{ij} = A_{ji}.$$

AN INTRODUCTION TO PROBABILITY

Basic Definitions

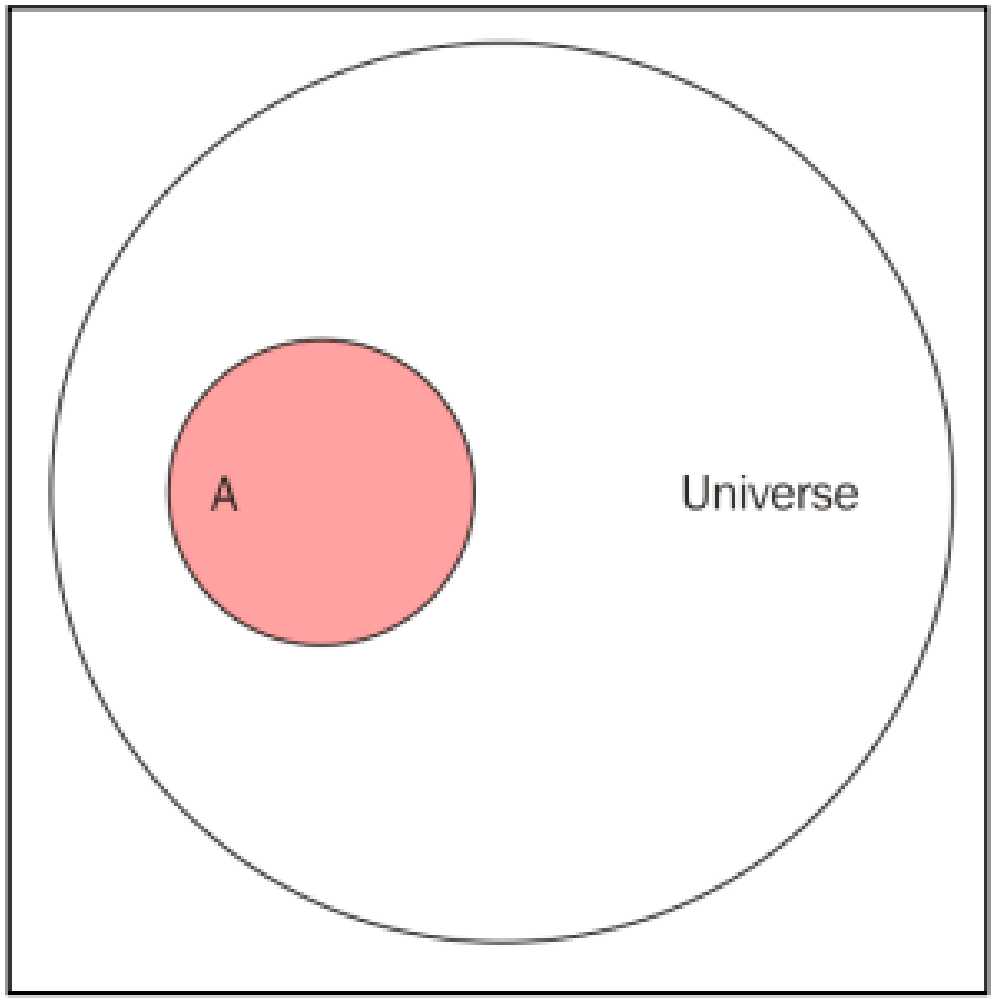
- **Procedure**
  - A procedure is an act that leads to a result.
- **Event**
  - An event is a collection of the outcomes of a procedure.
- **Simple Event**
- **Compound Event**

Basic Definitions

- **Sample Space**
  - The sample space of a procedure is the set of all possible simple events.
  - E.g. {HHH, HHT, HTT, HTH, TTT, TTH, THH, or THT}.
- The **probability** of an event represents the frequency, or chance, that the event will happen.

$$P(A) = \frac{\text{number of ways A occur}}{\text{size of sample space}}$$

Basic Definitions



Bayesian versus Frequentist

- In a Frequentist approach, the probability of an event is calculated through experimentation.
- It uses the past in order to predict the future chance of an event.

$$P(A) = \frac{\text{number of times A occurred}}{\text{number of times the procedure was repeated}}$$

Frequentist Approach

- **Example – marketing stats**
  - The rate of repeat visitors
- **Let's say, of the 1,458 unique visitors in the past week, 452 were repeat visitors. We can calculate this as follows:**

$$P(A)\ RF(A) = \frac{452}{1458} = .31$$

The law of large numbers

- For Example you want to pick the average number between 1 and 10.
- Let's design the experiment to be as follows:

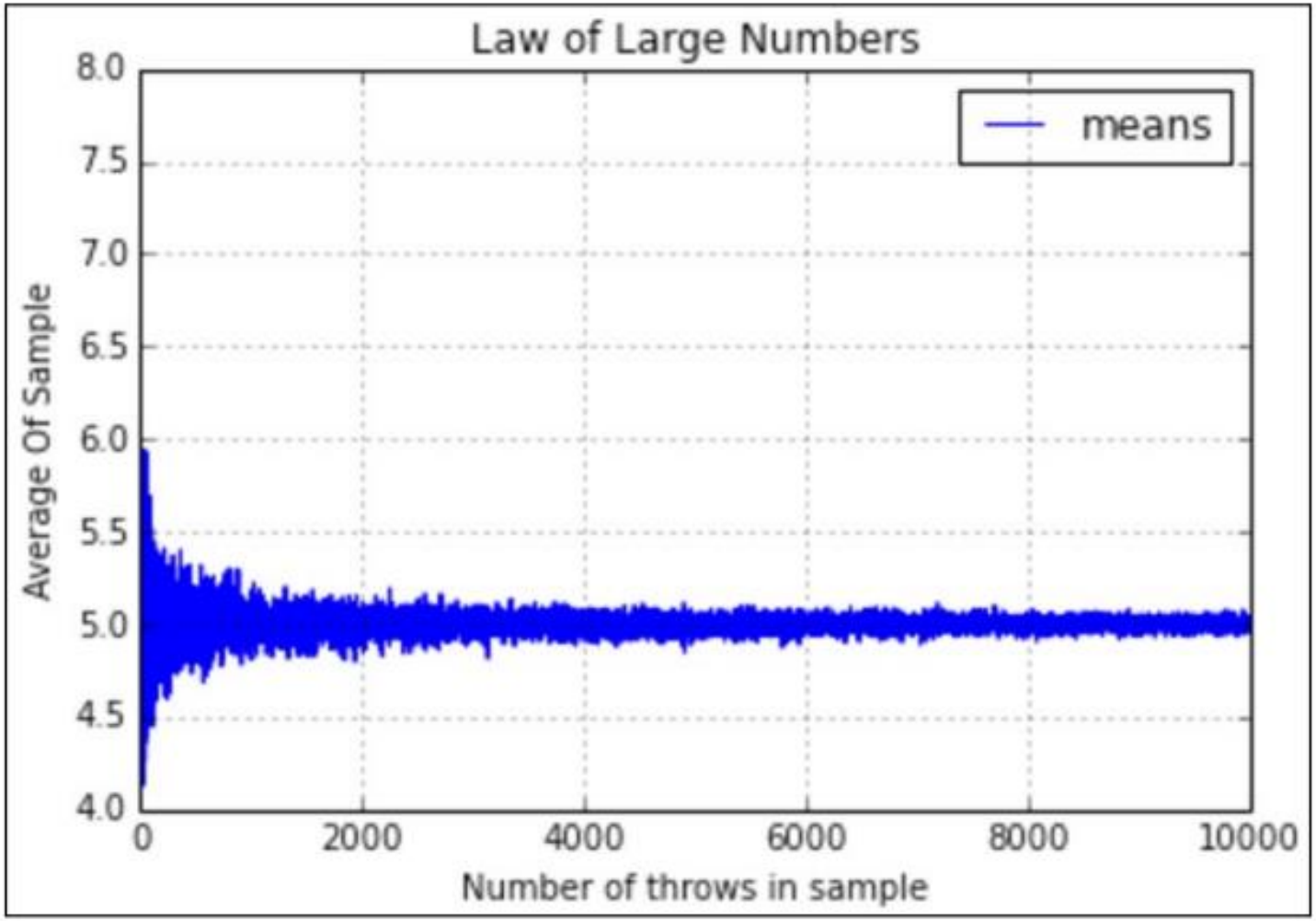
1. Pick a random number between 1 and 10 and find the average.
2. Pick two random numbers between 1 and 10 and find their average.
3. Pick three random numbers between 1 and 10 and find their average.
4. Pick 10,000 random numbers between 1 and 10 and find their average.
5. Graph the results

Let's take a look at the code:

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
%matplotlib inline
results = []
for n in range(1,10000):
    nums = np.random.randint(low=1,high=10, size=n) # choose n numbers
between 1 and 10
    mean = nums.mean()                              # find the average
of these numbers
    results.append(mean)                            # add the average
to a running list

# POP QUIZ: How large is the list results?
len(results) # 9999
# This was tricky because I took the range from 1 to 10000 and usually
we do from 0 to 10000
df = pd.DataFrame({ 'means' : results})
print df.head() # the averages in the beginning are all over the
place!
# means
# 9.0
# 5.0
# 6.0
# 4.5
# 4.0
print df.tail() # as n, our size of the sample size, increases, the
averages get closer to 5!
# means
# 4.998799
# 5.060924
# 4.990597
# 5.008802
# 4.979198
df.plot(title='Law of Large Numbers')
plt.xlabel("Number of throws in sample")
plt.ylabel("Average Of Sample")
```

The law of large numbers



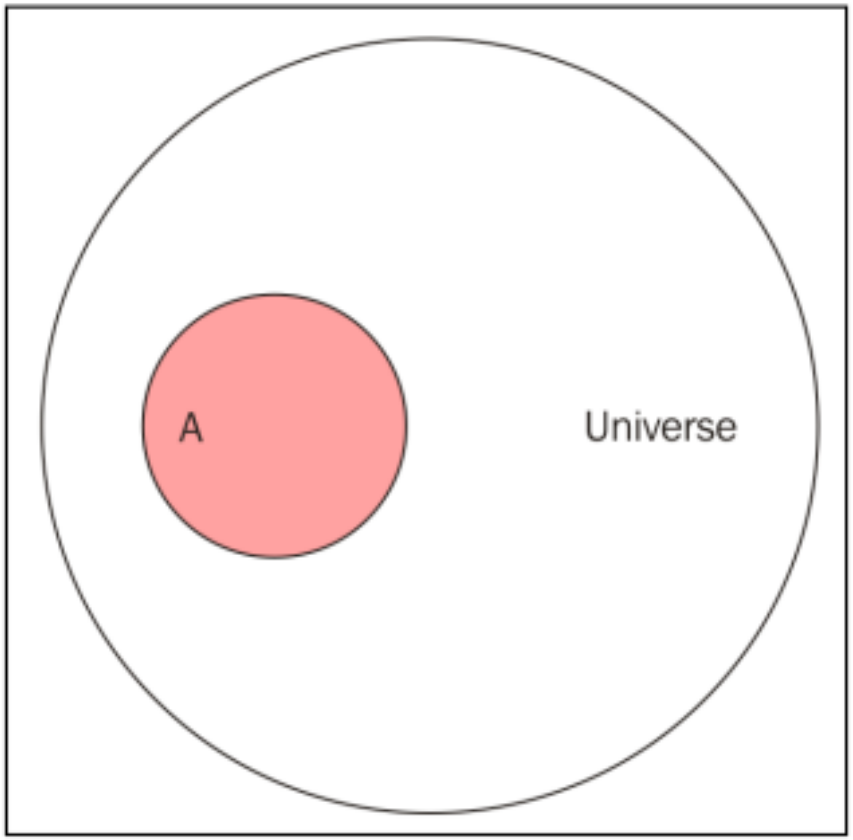
Compound events

Given events A and B:

- **The probability that A and B occur is $P(A \cap B)$ = P(A and B)**
- **The probability that either A or B occurs is P(A U B) = P(A or B)**
- **Let's say that our Universe is 100 people who showed up for an experiment, in which a new test for cancer is being developed:**

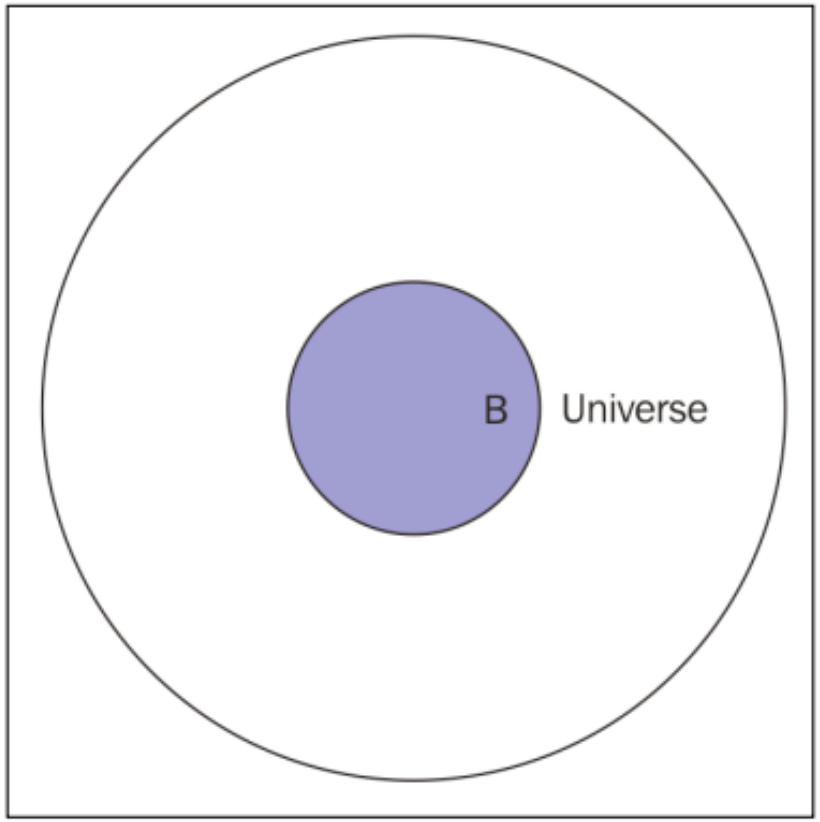
Compound events

The red circle, A, represents 25 people who actually have cancer.

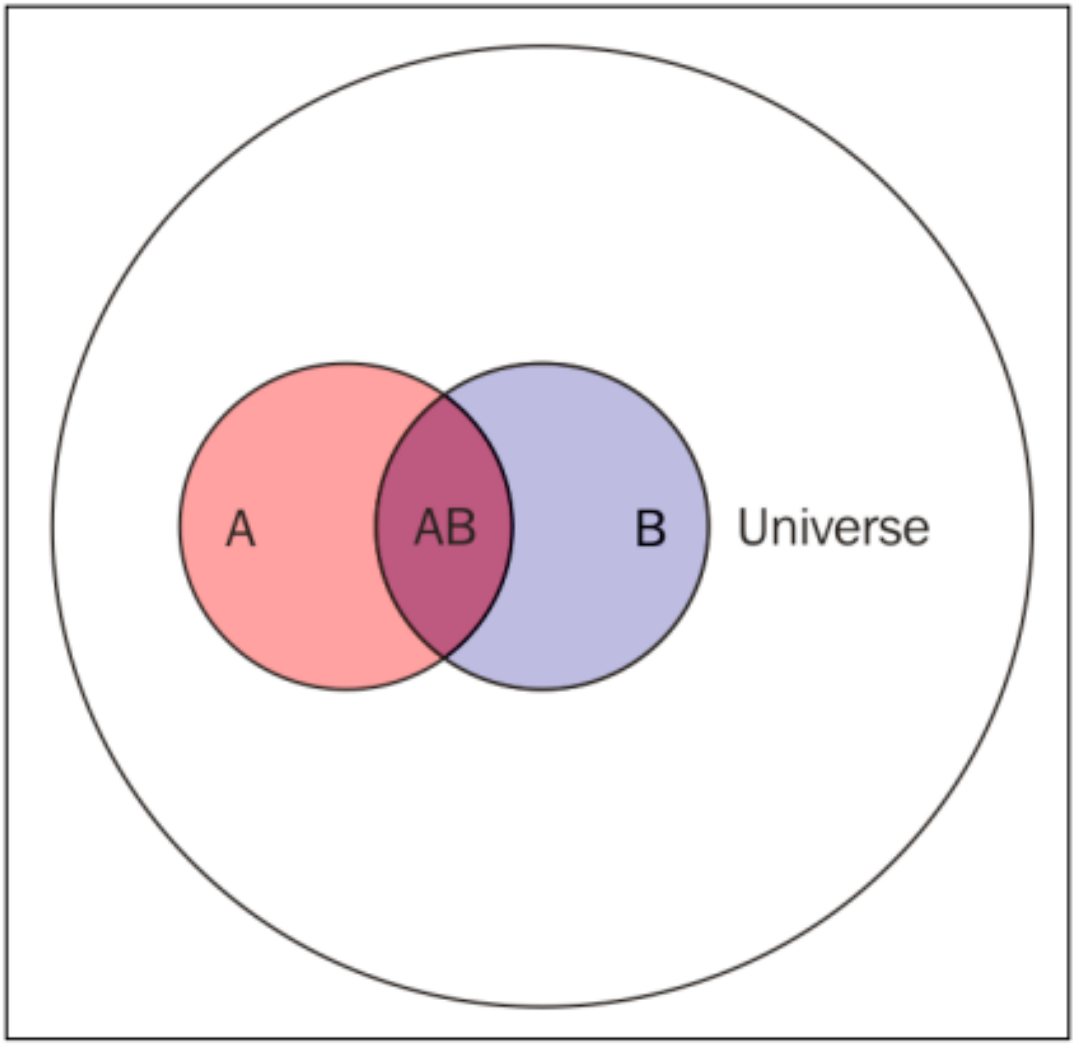


Compound events

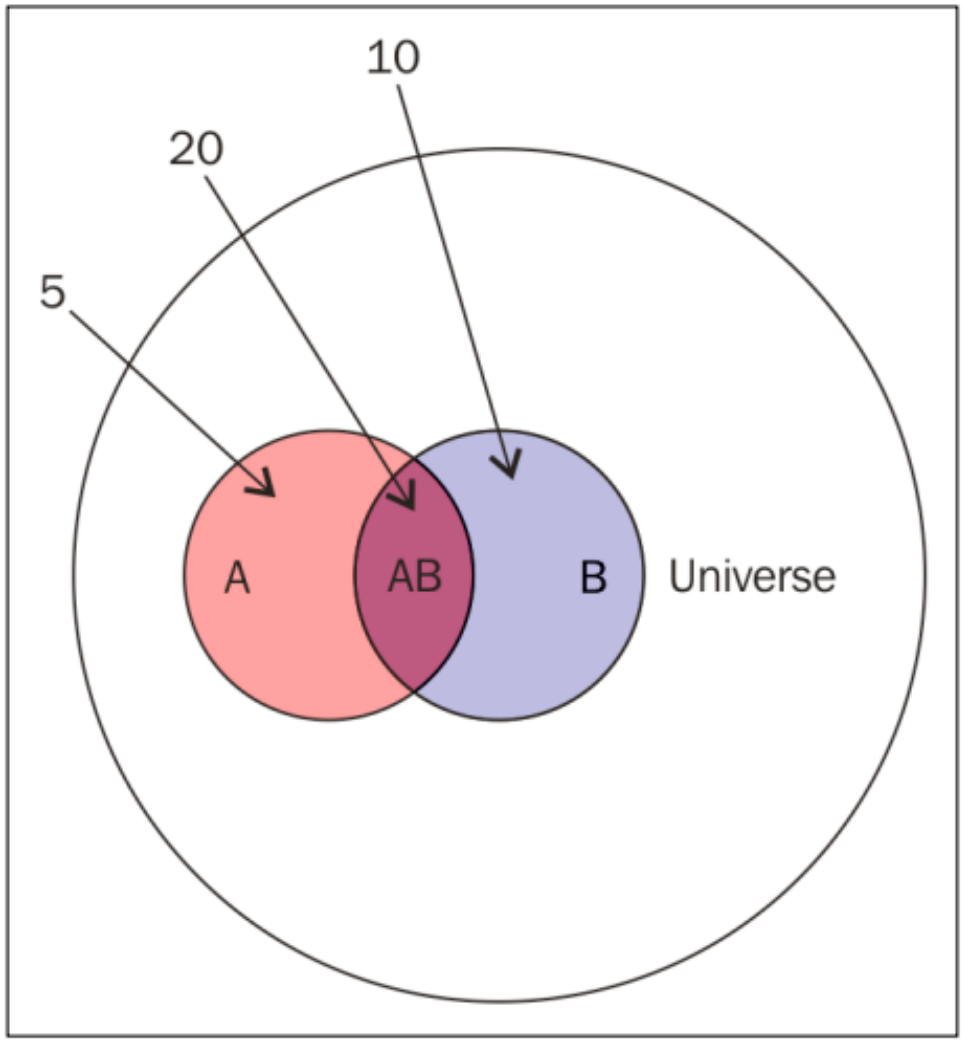
B shows 30 people for whom the test was positive (it claimed that they had cancer).



Compound events



Compound events

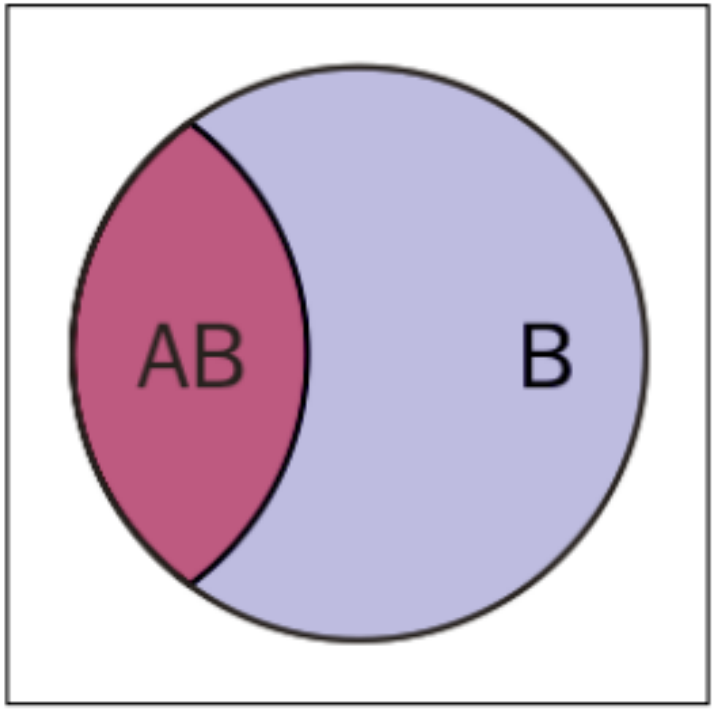


Compound events

All in all, we have people in the following four different classes:

- Pink: This refers to the people who have cancer and had a negative test outcome
- Purple (A intersect B): These people have cancer and had a positive test outcome
- Blue: This refers to the people with no cancer and a positive test outcome
- White: This refers to the people with no cancer and a negative test outcome

Conditional probability



P(A|B) = P(A and B) / P(B) = (20/100) / (30/100) = 20/30 = .66 = 66%

Conditional probability

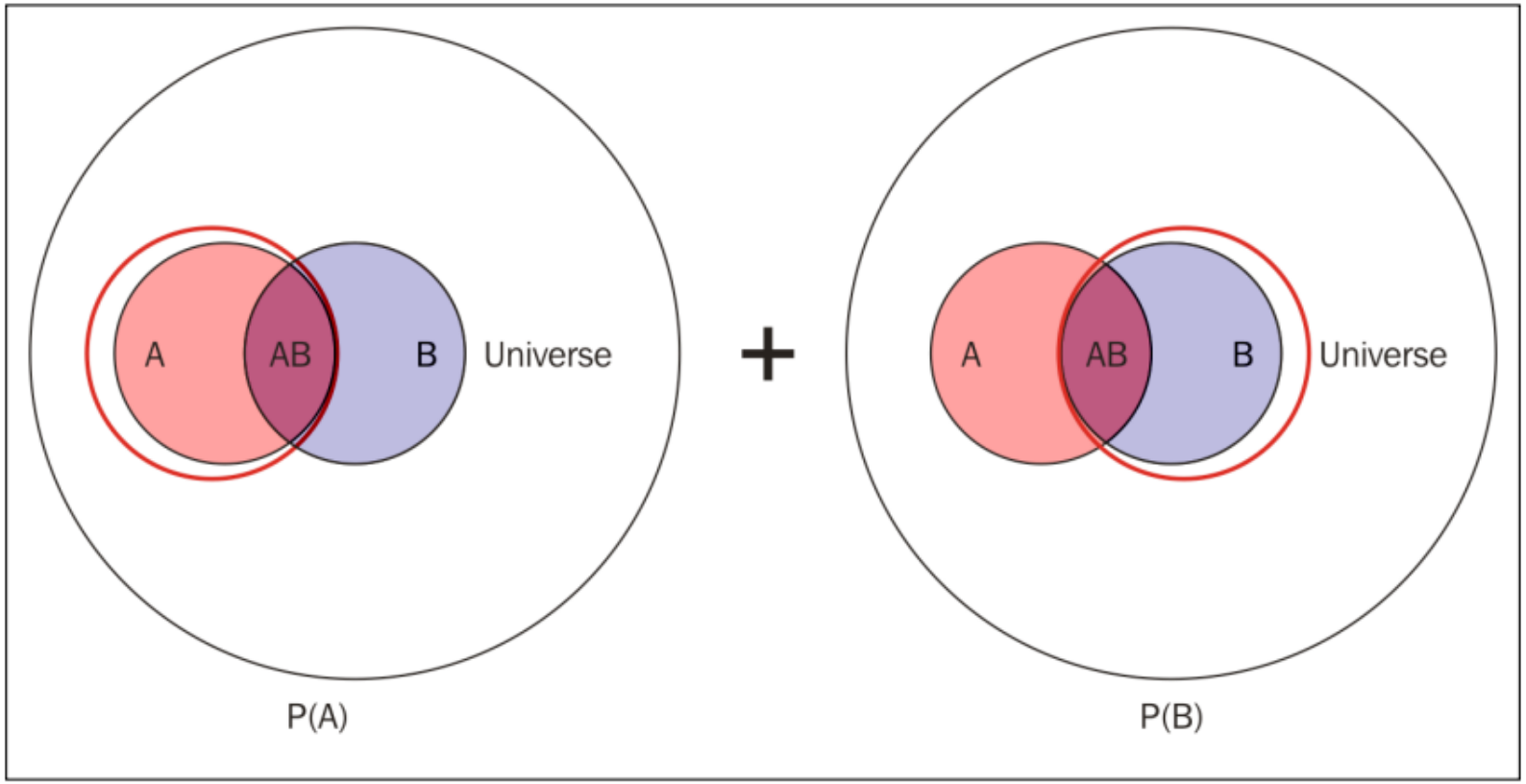
Conditional probability is used in

- Finance
- Insurance
- Science
- Machine learning

The rules of probability

The addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



The rules of probability

Mutual exclusivity

$A \cap B = \emptyset$

$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B)$

Examples:

Today is Saturday and today is Wednesday

I failed the exam and I passed the exam

None of these events can occur simultaneously.

The rules of probability

The multiplication rule

- Read Example - Principles of Data Science (page 99-100).
- The conditional probability is very important in the multiplication rule as it can drastically alter your answer.
- For example, of a randomly selected set of 10 people, 6 have iPhones and 4 have Androids. What is the probability that if I randomly select two people, they both will have iPhones?

The rules of probability

- A: This event shows the probability that I choose a person with an iPhone first
- B: This event shows the probability that I choose a person with an iPhone second
- **P(A and B): P(I choose a person with an iPhone and a person with an iPhone)**
- People with iPhones are 6 out of 10, so, I have a 6/10 = 3/5 = 0.6 chance of A.
- This means P(A) = 0.6.

The rules of probability

- The probability of choosing two should just be 0.6 * 0.6, right?
- A: This event shows the probability that I choose a person with an iPhone first
- B: This event shows the probability that I choose a person with an iPhone second
- P(A and B): P(I choose a person with an iPhone and a person with an iPhone)

The rules of probability

- $P(A \text{ and } B) = P(A) \cdot P(B|A)$ formula
- $P(A)$ is simple, right?
- People with iPhones are 6 out of 10, so, I have a 6/10 = 3/5 = 0.6 chance of A. This means $P(A) = 0.6$. Right?
- But wait! We only have 9 people left to choose our second person from, because one was taken away. So in our new transformed sample space, we have 9 people in total, 5 with iPhones and 4 with droids, making $P(B) = 5/9 = .555$.

Independence

Two events are independent if one event does not affect the outcome of the other, that is $P(B|A) = P(B)$ and $P(A|B) = P(A)$.

Some examples of independent events are as follows:

- It was raining in San Francisco, and a puppy was born in India
- Flip a coin and get heads and flip another coin and get tails

Complementary events

To calculate the probability of $\overline{A}$, use the following formula:

$$P(\overline{A}) = 1 - P(A)$$

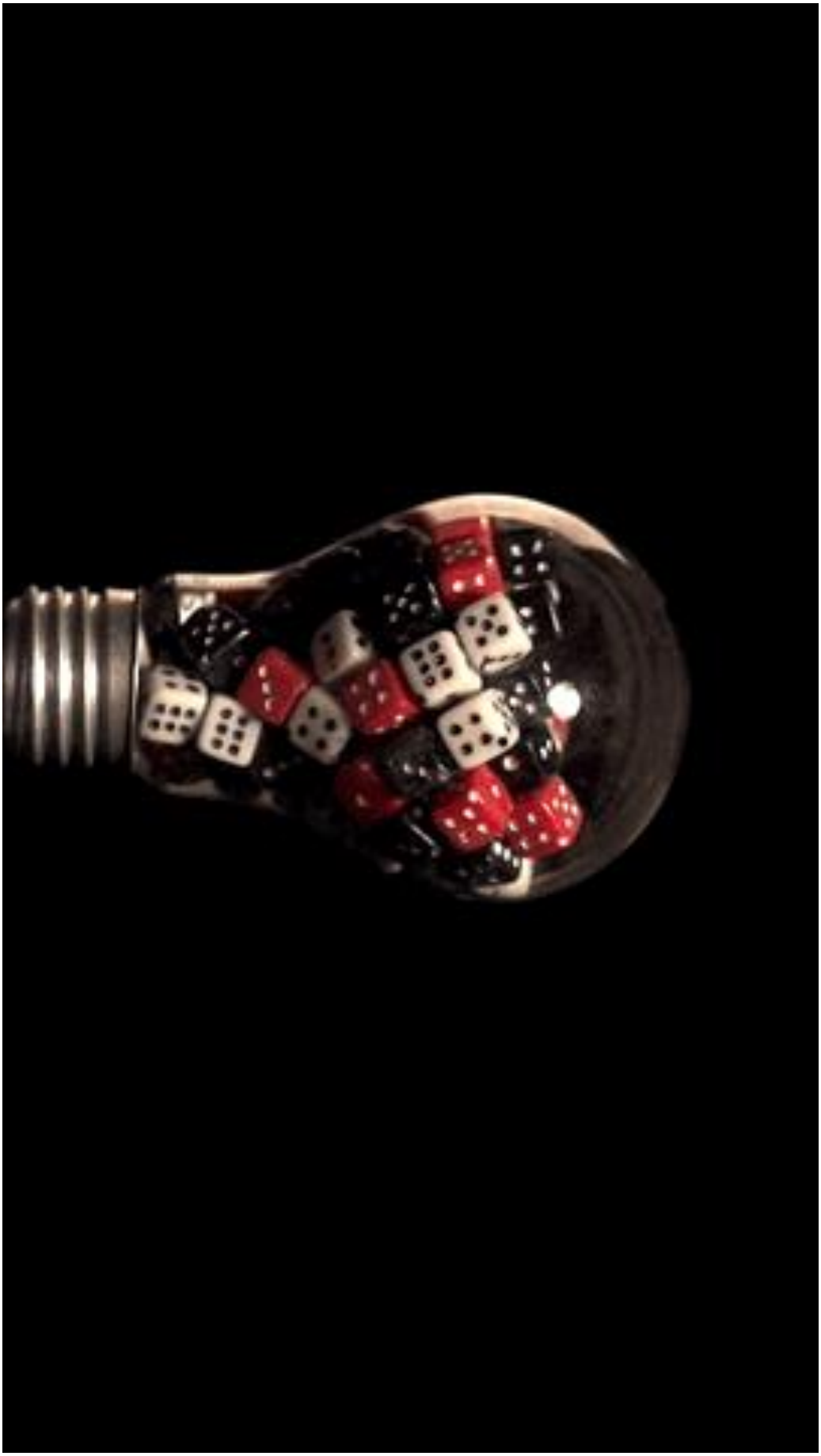
For example, a start-up team has three investor meetings coming up. We will have the following probabilities:

- 60% chance of getting money from the first meeting
- 15% chance of getting money from the second
- 45% chance of getting money from the third

Addition rule applies to mutually exclusive events

Multiplication rule applies to independent events

A bit deeper

Let's try some different numbers. Let's say 165 people walked in for the study. So, our n (sample size) is 165 people. All 165 people are given the test and asked if they have cancer (provided through various other means).

n=165	**Predicted: NO**	**Predicted: YES**
Actual: NO	50	10
Actual: YES	5	100